



Master in Computer Science
Software Engineering

Cloud Computing

Report 1 **Team Members**

Jordy Meye
Huizhe Su
Arihant Jain
Nizar

February 2026

Contents

1	Server Installation	2
1.1	Install Ubuntu 18.04 LTS in Amazon EC2	2
1.2	Update the system	2
1.3	Find the main characteristics of your instance from the commands below.	2
2	Installing Apache Web Server	7
2.4	Install the Apache 2 HTTP Server	7
2.5	Check the Status of Apache 2 and Give Its Version	7
2.6	Create a Home Page and Test with a Web Browser	8
2.7	Create a DNS Zone with CloudDNS. Give the CNAME and ALIAS Created	9
2.8	Check DNS Resolution with the dig Command	9
2.9	Test with a Web Browser	10
2.10	Create and Install a Self-Signed Certificate. Test with a Web Browser	10
2.11	Create and Install a Certified Certificate with ZeroSSL. Test with a Web Browser	11
2.12	Create and Install a Certified Certificate with Let's Encrypt and Certbot. Test with a Web Browser	13
2.13	Enable the Apache Rewrite Engine and Create the HTTP to HTTPS Redirect Rule. Test with a Web Browser	14
3	Creating a PHP Application	15
3.14	Install the PHP Module	15
3.15	Create a phpinfo.php Script and Identify the PHP Version	15
3.16	Create a meteo.php Script That Displays Weather Information Using OpenWeather	16

Exercise 1. Server Installation

Question 1. Install Ubuntu 18.04 LTS in Amazon EC2

After registering on AWS and setting up an EC2 instance, we have the following information page.

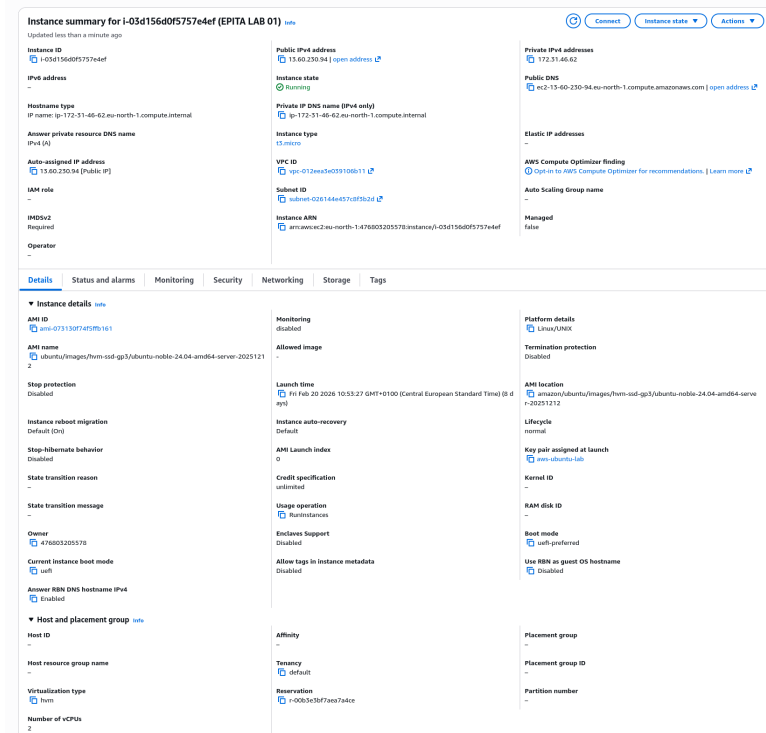


Figure 1: AWS EC instance information page (Huizhe Su)

It's recommended to use ssh to login to the remote server. First, we download our ssh key from our account, then we need to change the permission of the private key to be read-only by user only.

```
chmod 400 ./aws-ubuntu-lab.pem
```

For easy access our remote server, we can edit `~/.ssh/config` and add the following configuration.

```
Host aws_epita
  HostName ec2-13-60-230-94.eu-north-1.compute.amazonaws.com
  User ubuntu
  IdentityFile ~/.ssh/aws-ubuntu-lab.pem
```

Question 2. Update the system

We update the server using the following command:

```
sudo apt-get install update
sudp apt-get install upgrade
```

Question 3. Find the main characteristics of your instance from the commands below.

Instead of output the information by command line, we can use a new tool called `neofetch`. Install `neofetch` and run `neofetch`

```
sudo apt-get install neofetch
neofetch
```

Then we will get the information

```
[ip-172-31-] ~

User sessions running outdated binaries:
ubuntu @ session #4: sshd[1222,1295]
ubuntu @ user manager service: systemd[1227]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-46-62 ~> neofetch

.-/+00ssssss+/-.
`:+ssssssssssssssssssss+`
+ssssssssssssssssssssyyssss+
.oSSssssssssssssssssssdMMMMNysSSso.
/SSsssssssssshdmmNNmmymMMMMhSSssss/
+SSssssssshmydMMMMMMNdddddysSSssss+
/SSssssshNMMMyhhyyyhmmNNMMhSSssss/
.SSSssssdMMMNhSSssssssshNMMMdSSssss.
+SSshhhyNMMNysSSsssssssyNMMMySSssss+
ossyNMMMNyMMhSSssssssssshmmhSSssssso
ossyNMMMNyMMhSSssssssssshmmhSSssssso
+SSshhhyNMMNysSSsssssssyNMMMySSssss+
.SSSssssdMMMNhSSssssssshNMMMdSSssss.
/SSssssshNMMMyhhyyyhNMMNhSSssss/
+SSssssssdmydMMMMMMNdddddysSSssss+
/SSssssssshdmmNNmmymMMMMhSSssss/
.oSSssssssssssssssssssdMMMMNysSSso.
+SSssssssssssssssssssyyssss+
`:+ssssssssssssssssss+`
.-/+00ssssss+/-.

ubuntu@ip-172-31-46-62 ~>

ubuntu@ip-172-31-46-62
-----
OS: Ubuntu 24.04.4 LTS x86_64
Host: t3.micro
Kernel: 6.14.0-1018-aws
Uptime: 16 mins
Packages: 773 (dpkg), 4 (snap)
Shell: bash 5.2.21
Resolution: 800x600
Terminal: /dev/pts/0
CPU: Intel Xeon Platinum 8175M (2) @
GPU: 00:03.0 Amazon.com, Inc. Device
Memory: 201MiB / 914MiB
```

Figure 2: Neofetch and system information (Huizhe Su)

For other information, we selected to show `ssh` logs, network interfaces, detailed CPU information, installed packages, groups, `passwd`, devices, processes, fire wall status, hardware space.

```
journalctl -u ssh # log for ssh
ifconfig #network interfaces
cat /proc/cpuinfo # cpu info
dpkg -l | wc -l # calculate installed packages
# we use getent instead of cat for accessing system database
getent group # groups
getent passwd # passwd
lsusb # list usb devices
lspci # list peripheral devices
top # snapshot of current processes
ps x # list all processes
sudo ufw status verbose # show firewall status
df -l --si # hardware spaces
```

```
ubuntu@ip-172-31-46-62 ~> journalctl -u ssh
Feb 20 10:01:56 ip-172-31-46-62 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 20 10:01:56 ip-172-31-46-62 sshd[1044]: Server listening on 0.0.0.0 port 22.
Feb 20 10:01:56 ip-172-31-46-62 sshd[1044]: Server listening on :: port 22.
Feb 20 10:01:56 ip-172-31-46-62 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 20 10:01:57 ip-172-31-46-62 sshd[1046]: Connection closed by 163.5.3.41 port 5034 (preauth)
Feb 20 10:02:04 ip-172-31-46-62 sshd[1048]: Connection closed by authenticating user ubuntu 163.5.3.41 port 8276 (preauth)
Feb 20 10:02:30 ip-172-31-46-62 sshd[1050]: Accepted publickey for ubuntu from 163.5.3.41 port 52284 ssh2: E025519 SHA256:bYcPE818hQLDptoad7Ck1cpqjY3T6p
Feb 20 10:02:30 ip-172-31-46-62 sshd[1050]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
Feb 20 10:04:57 ip-172-31-46-62 sshd[1222]: Accepted publickey for ubuntu from 163.5.3.41 port 13218 ssh2: E025519 SHA256:bYcPE818hQLDptoad7Ck1cpqjY3T6p
Feb 20 10:04:57 ip-172-31-46-62 sshd[1222]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
Feb 20 10:08:40 ip-172-31-46-62 sshd[1044]: Received signal 15; terminating.
Feb 20 10:08:40 ip-172-31-46-62 systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Feb 20 10:08:40 ip-172-31-46-62 systemd[1]: ssh.service: Deactivated successfully.
Feb 20 10:08:40 ip-172-31-46-62 systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
Feb 20 10:08:40 ip-172-31-46-62 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 20 10:08:40 ip-172-31-46-62 sshd[9257]: Server listening on 0.0.0.0 port 22.
Feb 20 10:08:40 ip-172-31-46-62 sshd[9257]: Server listening on :: port 22.
Feb 20 10:08:40 ip-172-31-46-62 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Figure 3: User level logs of ssh service(Huizhe Su)

```

no VM guests are running outside hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-46-62 ~$ ifconfig
ens5: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 172.31.46.62 netmask 255.255.240.0 broadcast 172.31.47.255
    inet6 fe80::895:18ff:fe83:fd07 prefixlen 64 scopeid 0x20<link>
    ether 0a:95:18:83:fd:07 txqueuelen 1000 (Ethernet)
    RX packets 143362 bytes 191407568 (191.4 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 17905 bytes 2096180 (2.0 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 250 bytes 28379 (28.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 250 bytes 28379 (28.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

Figure 4: Network interfaces (Huizhe Su)

```

ubuntu@ip-172-31-46-62 ~$ cat /proc/cpuinfo
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 85
model name     : Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz
stepping       : 4
microcode      : 0x2007000
cpu MHz        : 2499.998
cache size     : 33792 KB
physical id    : 0
siblings       : 2
core id        : 0
cpu cores      : 1
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm co
instant_tsc_rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq sse3 fma cx16 pcid sse4.1 sse4.2 x2apic movbe popcnt tsc_deadline_timer
r_uses_xsave avx f16c rdrand hypervisorlahf_lm abm 3dnowprefetch cpuid_fault pti fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid mpx avx512f avx512d
q_rdtseed adx snap clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaVES idA arat pku ospke
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapsgs itlb_multihit mmio_stale_data retbleed gds bhi its
bogomips       : 4999.99
clflush size   : 64
cache_alignmen : 64
address sizes   : 46 bits physical, 48 bits virtual
power managemen:

processor       : 1
vendor_id      : GenuineIntel
cpu family     : 6
model          : 85
model name     : Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz
stepping       : 4
microcode      : 0x2007000
cpu MHz        : 2499.998
cache size     : 33792 KB
physical id    : 0
siblings       : 2
core id        : 0
cpu cores      : 1
apicid         : 1
initial apicid : 1
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm co
instant_tsc_rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq sse3 fma cx16 pcid sse4.1 sse4.2 x2apic movbe popcnt tsc_deadline_timer
r_uses_xsave avx f16c rdrand hypervisorlahf_lm abm 3dnowprefetch cpuid_fault pti fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid mpx avx512f avx512d
q_rdtseed adx snap clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaVES idA arat pku ospke
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapsgs itlb_multihit mmio_stale_data retbleed gds bhi its
bogomips       : 4999.99
clflush size   : 64
cache_alignmen : 64
address sizes   : 46 bits physical, 48 bits virtual
power managemen:

```

Figure 5: Detailed CPU information (Huizhe Su)

```

ubuntu@ip-172-31-46-62 ~$ dpkg -l | wc -l
778

```

Figure 6: There are 778 installed packages (Huizhe Su)

```

ubuntu@ip-172-31-46-62 ~$ getent group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,ubuntu
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:ubuntu
floppy:x:25:
tape:x:26:
sudo:x:27:ubuntu
audio:x:29:
dip:x:30:ubuntu
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
shadow:x:42:
utmp:x:43:
video:x:44:
sasl:x:45:
plugdev:x:46:
staff:x:50:
games:x:60:
users:x:100:
nogroup:x:65534:
systemd-journal:x:999:
systemd-network:x:998:
crontab:x:997:
systemd-timesync:x:996:
input:x:995:
sgx:x:994:
kvm:x:993:
render:x:992:
messagebus:x:101:
syslog:x:102:
systemd-resolve:x:991:
uudd:x:103:
tss:x:104:
lxd:x:105:ubuntu
ssh:x:106:
rdma:x:107:
tcpdump:x:108:
landscape:x:109:
fwupd-refresh:x:990:
polkitd:x:989:
admin:x:110:
netdev:x:111:
chrony:x:112:
ubuntu:x:1000:
ubuntu@ip-172-31-46-62 ~$

```

Figure 7: Group information

```

ubuntu@ip-172-31-46-62 ~$ getent passwd
root:x:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,:/usr/lib/dhcpcd/bin/false
messagebus:x:101:101:/nonexistent:/usr/sbin/nologin
syslog:x:102:102:/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/usr/sbin/nologin
uidd:x:103:103:/run/uidd:/usr/sbin/nologin
tss:x:104:104:TPM software stack,,:/var/lib/tpm/bin/false
sshd:x:105:65534:/run/ssh:/usr/sbin/nologin
pollinate:x:106:1:/var/cache/pollinate/bin/false
tcpdump:x:107:108:/nonexistent:/usr/sbin/nologin
landscape:x:108:109:/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/usr/sbin/nologin
ec2-instance-connect:x:109:65534:/nonexistent:/usr/sbin/nologin
chrony:x:110:112:Chrony daemon,,:/var/lib/chrony:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
ubuntu@ip-172-31-46-62 ~$ getent group

```

Figure 8: User information

```

ubuntu@ip-172-31-46-62 ~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           914Mi       397Mi       197Mi       2.7Mi       493Mi       516Mi
Swap:            0B           0B           0B

ubuntu@ip-172-31-46-62 ~$ lsusb
ubuntu@ip-172-31-46-62 ~$ [1] ls pci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma]
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.3 Non-VGA unclassified device: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:03.0 VGA compatible controller: Amazon.com, Inc. Device 1111
00:04.0 Non-Volatile memory controller: Amazon.com, Inc. NVMe EBS Controller
00:05.0 Ethernet controller: Amazon.com, Inc. Elastic Network Adapter (ENA)
ubuntu@ip-172-31-46-62 ~$

```

Figure 9: USB information and peripheral information (Huizhe Su)

```
top - 10:27:00 up 33 min, 1 user, load average: 0.00, 0.00, 0.02
Tasks: 108 total, 1 running, 107 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 914.2 total, 177.6 free, 383.7 used, 527.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 530.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
1295 ubuntu  20   0   14992   4816   2852  S   0.3   0.5   0:01.85 sshd
 1 root    20   0   22592  14028   9704  S   0.0   1.5   0:06.04 systemd
 2 root    20   0     0     0     0  S   0.0   0.0   0:00.00 kthreadd
 3 root    20   0     0     0     0  S   0.0   0.0   0:00.00 pool_workqueue_release
 4 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-rcu_gp
 5 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-sync_wq
 6 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
 7 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
 8 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-netns
10 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
13 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
14 root    20   0     0     0     0  I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
15 root    20   0     0     0     0  I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
16 root    20   0     0     0     0  S   0.0   0.0   0:00.07 ksoftirqd/0
17 root    20   0     0     0     0  I   0.0   0.0   0:00.30 rcu_sched
18 root    20   0     0     0     0  S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
19 root    20   0     0     0     0  S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
20 root    rt    0     0     0     0  S   0.0   0.0   0:00.00 migration/0
21 root   -51   0     0     0     0  S   0.0   0.0   0:00.00 idle_inject/0
22 root    20   0     0     0     0  S   0.0   0.0   0:00.00 cpuhp/0
23 root    20   0     0     0     0  S   0.0   0.0   0:00.00 cpuhp/1
24 root   -51   0     0     0     0  S   0.0   0.0   0:00.00 idle_inject/1
25 root    rt    0     0     0     0  S   0.0   0.0   0:00.07 migration/1
26 root    20   0     0     0     0  S   0.0   0.0   0:00.08 ksoftirqd/1
27 root    20   0     0     0     0  I   0.0   0.0   0:00.23 kworker/1:0-events
28 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
29 root    20   0     0     0     0  S   0.0   0.0   0:00.00 kdevtmpfs
30 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
31 root    20   0     0     0     0  S   0.0   0.0   0:00.00 kauditd
32 root    20   0     0     0     0  S   0.0   0.0   0:00.00 khungtaskd
33 root    20   0     0     0     0  I   0.0   0.0   0:00.25 kworker/u8:1-flush-259:0
34 root    20   0     0     0     0  S   0.0   0.0   0:00.00 dmcc_reaper
35 root    20   0     0     0     0  I   0.0   0.0   0:00.27 kworker/u8:2-events_power_efficient
36 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-writeback
37 root    20   0     0     0     0  S   0.0   0.0   0:00.06 kcompactd0
38 root    25   5     0     0     0  S   0.0   0.0   0:00.00 ksmd
39 root    39  19     0     0     0  S   0.0   0.0   0:00.00 khugepaged
40 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-kintegrityd
41 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-kblockd
42 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-blkg_punt_bio
43 root   -51   0     0     0     0  S   0.0   0.0   0:00.00 irq/9-acpi
45 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-tpm_dev_wq
46 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-ata_sff
47 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-md
48 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-md_bitmap
49 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-edac-poller
50 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-devfreq_wq
51 root   -51   0     0     0     0  S   0.0   0.0   0:00.00 watchdogd
52 root    0 -20   0     0     0  I   0.0   0.0   0:00.01 kworker/1:1H-kblockd
53 root    20   0     0     0     0  S   0.0   0.0   0:00.40 kswapd0
54 root    20   0     0     0     0  S   0.0   0.0   0:00.00 ecryptfs-kthread
55 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-kthrotld
56 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-acpi_thermal_pm
57 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-nvme-wq
58 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-nvme-reset-wq
59 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-nvme-delete-wq
60 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-nvme-auth-wq
62 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-ml
63 root    0 -20   0     0     0  I   0.0   0.0   0:00.01 kworker/0:1H-kblockd
64 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-ipv6_addrconf
71 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-kstrp
73 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/u9:0
86 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-charger_manager
87 root    20   0     0     0     0  S   0.0   0.0   0:00.42 jbd2/nvme0n1p1-8
88 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-ext4-rsv-conversion
163 root   0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-kmpathd
165 root   0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-kmpath_handlerd
251 root   0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-cryptd
252 root   0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-ena
288 root    20   0     0     0     0  S   0.0   0.0   0:00.00 jbd2/nvme0n1p16-8
289 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-ext4-rsv-conversion
593 message+ 20   0   10136   5028   3848  S   0.5   0.5   0:00.76 dbus-daemon
601 root    20   0   32416  20388  10140  S   0.2   0.2   0:00.08 networkd-dispat
616 root    20   0   18168   8608   7520  S   0.0   0.9   0:00.21 systemd-logind
743 root    20   0   6148   2220   2076  S   0.0   0.2   0:00.00 agetty
764 root    20   0  109996  20816  11360  S   0.2   2.2   0:00.07 unattended-upgr
807 root    20   0   6104   2104   1968  S   0.0   0.2   0:00.00 agetty
965 root    20   0  1830624  17328   8584  S   0.0   1.9   0:01.34 amazon-ssm-agent
1207 root    0 -20   0     0     0  I   0.0   0.0   0:00.00 kworker/R-tls-strp
1222 root    20   0   14736   6836   5140  S   0.0   0.7   0:00.01 sshd
1227 ubuntu  20   0   19912  11112   9376  S   0.0   1.2   0:00.16 systemd
1228 ubuntu  20   0   21152   3520   1792  S   0.0   0.4   0:00.00 (sd-pam)
1296 ubuntu  20   0   9060   4600   2896  S   0.0   0.5   0:00.03 bash
```

Figure 10: current processes (using top) (Huizhe Su)

```
ubuntu@ip-172-31-46-62 ~$ ps x
  PID TTY          STAT TIME COMMAND
 1227 ?           Ss    0:00 /usr/lib/systemd/systemd --user --deserialize=12
 1228 ?           S      0:00 (sd-pam)
 1295 ?           S      0:01 sshd: ubuntu@pts/0
 1296 pts/0       Ss    0:00 -bash
 9546 pts/0       Sl    0:01 fish
12499 pts/0       R+    0:00 ps x
```

Figure 11: current processes (using ps) (Huizhe Su)

```
ubuntu@ip-172-31-46-62 ~ [1]> sudo ufw status verbose
Status: inactive
ubuntu@ip-172-31-46-62 ~$
```

Figure 12: Firewall status (closed) (Huizhe Su)

```
ubuntu@ip-172-31-46-62 ~> runlevel
N 5
ubuntu@ip-172-31-46-62 ~> df -l --si
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        7.3G  2.6G  4.7G  36% /
tmpfs            480M    0  480M   0% /dev/shm
tmpfs            192M  914k  191M   1% /run
tmpfs            5.3M    0   5.3M   0% /run/lock
efivarfs         132k   3.7k   123k   3% /sys/firmware/efi/efivars
/dev/nvme0n1p16  924M   94M  766M  11% /boot
/dev/nvme0n1p15  110M   6.4M  103M   6% /boot/efi
tmpfs            96M    13k   96M   1% /run/user/1000
ubuntu@ip-172-31-46-62 ~> []
```

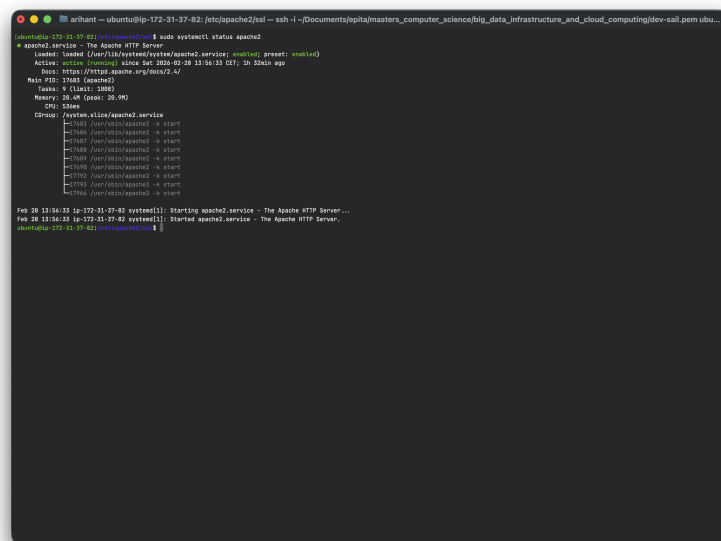
Figure 13: Free space on disk (Huizhe Su)

Exercise 2. Installing Apache Web Server

Question 4. Install the Apache 2 HTTP Server

To serve web content from our EC2 instance, we install the Apache2 web server and enable the service so that it starts automatically on boot.

```
sudo apt install apache2
sudo systemctl start apache2
sudo systemctl enable apache2
```



```
arihant ~ ubuntu@ip-172-31-37-82: /etc/apache2$ psal - ssh -l -Documents/apla/masters_computer_science/big_data_infrastructure_and_cloud_computing/dev-sal pem ubu...
ubuntu@ip-172-31-37-82: /etc/apache2$ sudo systemctl status apache2
+ apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2024-02-28 13:56:53 CEST; 1h 20min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 17683 (apache2)
     Tasks: 9 (limit: 65535)
    Memory: 28.4M (peak: 32.0M)
       CGroup: /system.slice/apache2.service
               └─17683 /usr/sbin/apache2 -k start
               └─17684 /usr/sbin/apache2 -k start
               └─17687 /usr/sbin/apache2 -k start
               └─17688 /usr/sbin/apache2 -k start
               └─17689 /usr/sbin/apache2 -k start
               └─17690 /usr/sbin/apache2 -k start
               └─17702 /usr/sbin/apache2 -k start
               └─17703 /usr/sbin/apache2 -k start
               └─17704 /usr/sbin/apache2 -k start

Feb 28 13:56:53 ip-172-31-37-82 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Feb 28 13:56:53 ip-172-31-37-82 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-37-82: /etc/apache2$
```

Figure 14: Apache2 service running successfully (Arihant Jain)

Question 5. Check the Status of Apache 2 and Give Its Version

We can verify that Apache is running correctly and check its version:

```
sudo systemctl status apache2
apache2 -v
```

The `systemctl status` command confirms that the service is **active (running)**. The `apache2 -v` command outputs the installed version:

```
Server version: Apache/2.4.58 (Ubuntu)
Server built:   2025-12-09T15:50:28
```

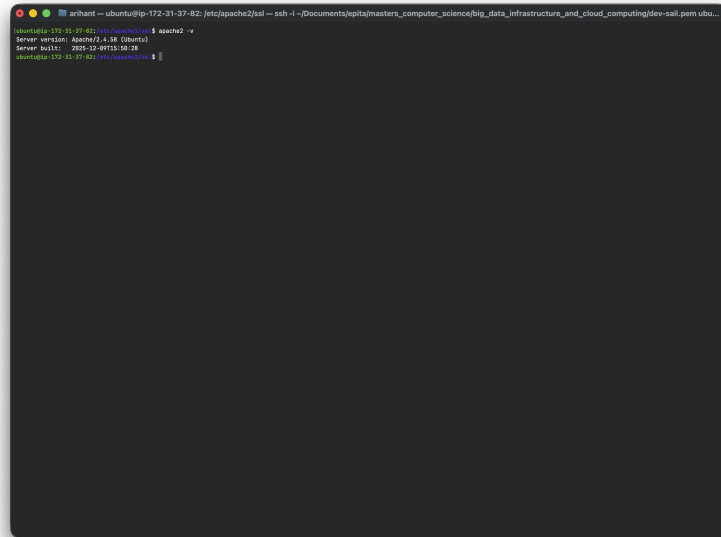


Figure 15: Apache2 version output (Arihant Jain)

Question 6. Create a Home Page and Test with a Web Browser

With Apache running, the default page is accessible at:

<http://ec2-35-180-10-25.eu-west-3.compute.amazonaws.com>

This serves the default Apache2 Ubuntu page located at `/var/www/html/index.html`. We then edited this file to serve our own content:

```
sudo nano /var/www/html/index.html
```

We replaced the default content with a simple custom page:

```
<html>
  <head>
    <title>My Server</title>
  </head>
  <body>
    Bienvenue sur mon serveur
  </body>
</html>
```

After saving the file, refreshing the browser shows our custom page.

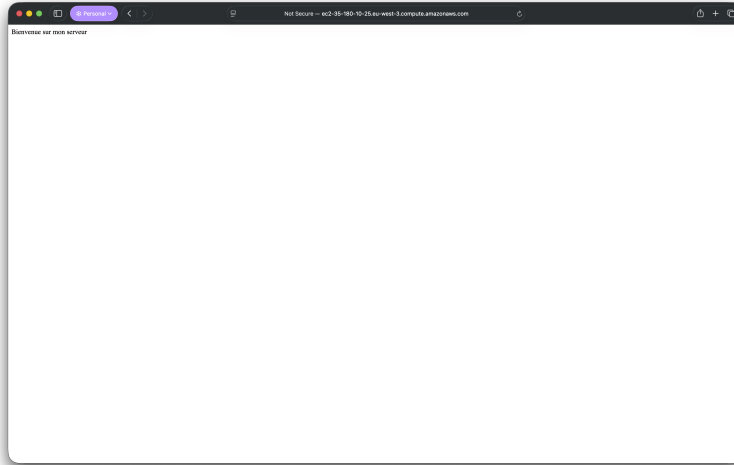


Figure 16: Custom home page: “Bienvenue sur mon serveur” (Arihant Jain)

Question 7. Create a DNS Zone with CloudDNS. Give the CNAME and ALIAS Created

To access our server using a custom domain name instead of the long EC2 public DNS, we set up a free DNS zone on CloudDNS:

1. Navigate to <https://www.cloudns.net>
2. Go to **Create a Zone** → **Free Zone**
3. Choose the name **arihant-dev-sail** with the suffix **.cloud-ip.cc**

We then created the following DNS records:

- **CNAME record:** **arihant-dev-sail.cloud-ip.cc** → **ec2-35-180-10-25.eu-west-3.compute.amazonaws.com**
- **A record (ALIAS):** **arihant-dev-sail.cloud-ip.cc** → **35.180.10.25**

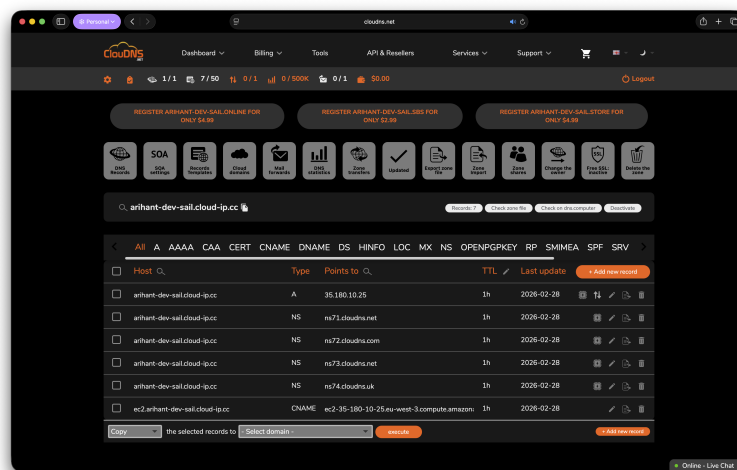


Figure 17: CloudDNS zone configuration with CNAME and A record (Arihant Jain)

Question 8. Check DNS Resolution with the dig Command

After DNS propagation, we verify that the domain resolves correctly using the **dig** command:

```
dig +short ec2.arihant-dev-sail.cloud-ip.cc
```

The output confirms the A record resolving to our EC2 public IPv4 address 35.180.10.25, confirming that the DNS zone is properly configured.

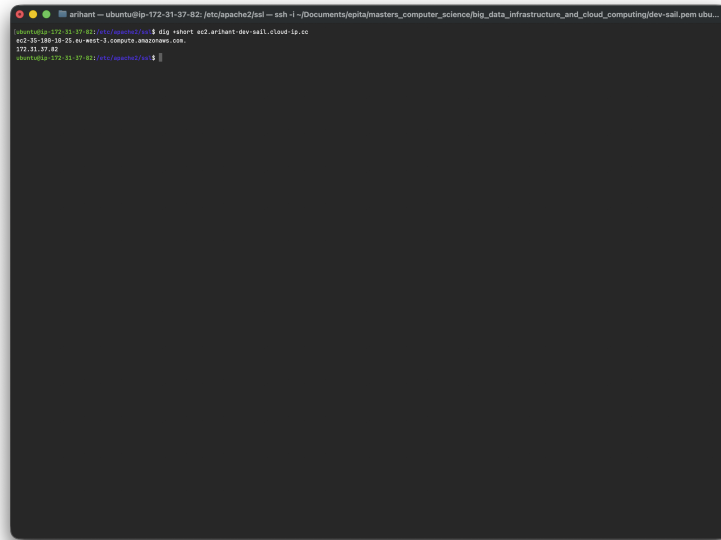


Figure 18: DNS resolution verified with dig (Arihant Jain)

Question 9. Test with a Web Browser

After DNS propagation, the server is now accessible via our custom domain at:

`http://arihant-dev-sail.cloud-ip.cc`

The browser displays our custom “Bienvenue sur mon serveur” page, confirming that the DNS is correctly pointing to our EC2 instance. However, the connection is not secured — the site is served over plain HTTP, as shown by the browser warning.

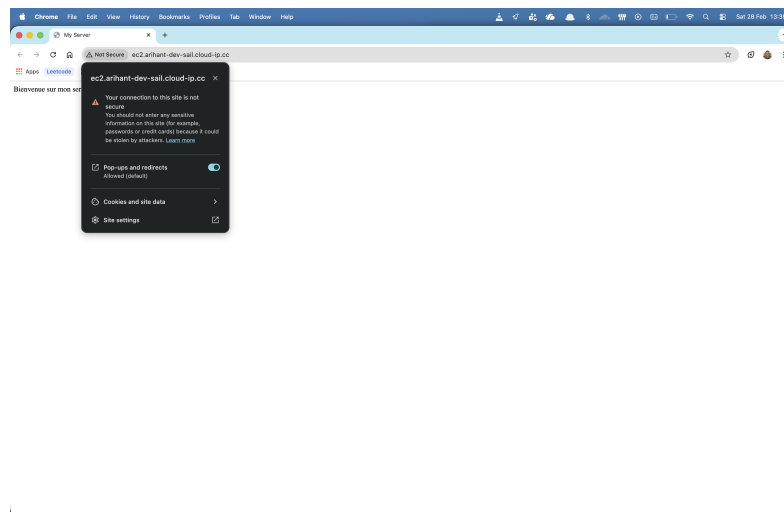


Figure 19: Site accessible via custom domain over HTTP (Arihant Jain)

Question 10. Create and Install a Self-Signed Certificate. Test with a Web Browser

As a first step toward securing the site, we generated a self-signed SSL certificate. We enabled the SSL module in Apache and configured it:

```
sudo a2enmod ssl
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
```

```

-keyout /etc/ssl/private/apache-selfsigned.key \
-out /etc/ssl/certs/apache-selfsigned.crt
sudo systemctl restart apache2

```

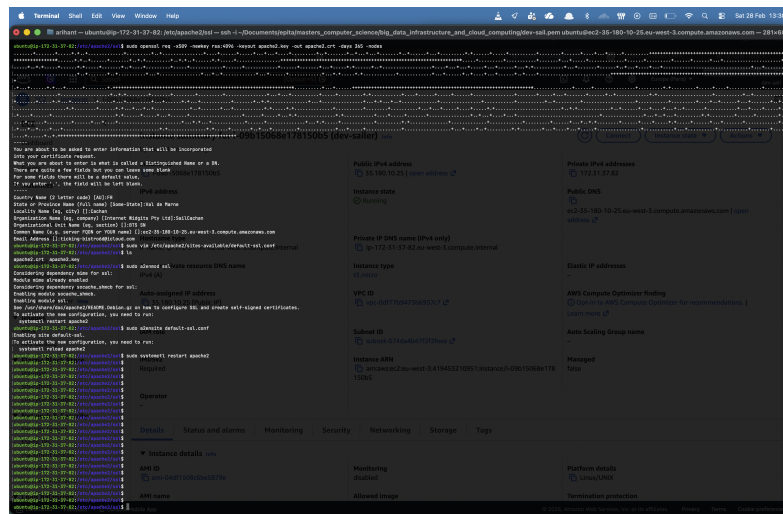


Figure 20: Generating a self-signed SSL certificate (Arihant Jain)

While this enables HTTPS, the browser displays a security warning because the certificate is not issued by a trusted Certificate Authority (CA). The connection is encrypted, but the browser cannot verify the server's identity.

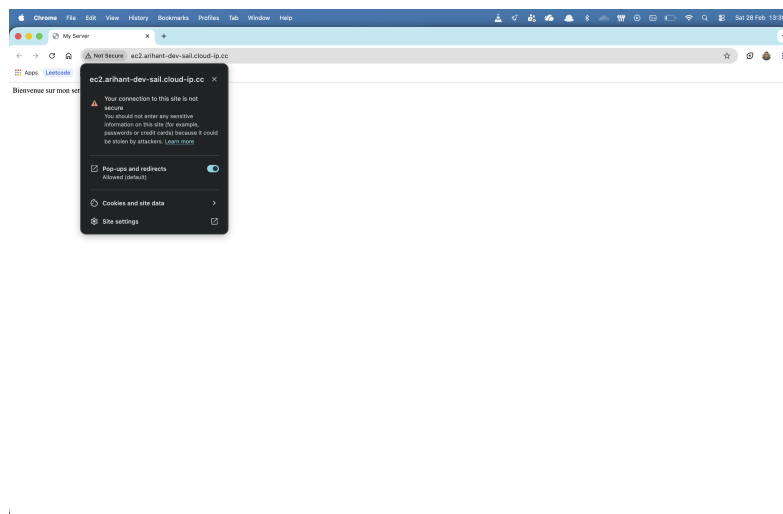


Figure 21: Browser warning for self-signed certificate (Arihant Jain)

Question 11. Create and Install a Certified Certificate with ZeroSSL. Test with a Web Browser

To get a ZeroSSL certificate, we need to register an account on ZeroSSL and request the certificate on the website. After that, we need to use CNAME verification to ensure we can install our certificate by adding the url for verification to our DNS Zone.

All A AAAA CAA CERT CNAME DNAME DS HINFO LOC MX NS OPENPGPKEY RP SMIMEA SPF SRV						
☐	Host	Type	Points to	TTL	Last update	+ Add new record
☐	marija-epita-lab.cloud-ip.cc	NS	ns71.cloudns.net	1h	2026-02-20	edit copy delete
☐	marija-epita-lab.cloud-ip.cc	NS	ns72.cloudns.com	1h	2026-02-20	edit copy delete
☐	marija-epita-lab.cloud-ip.cc	NS	ns73.cloudns.net	1h	2026-02-20	edit copy delete
☐	marija-epita-lab.cloud-ip.cc	NS	ns74.cloudns.uk	1h	2026-02-20	edit copy delete
☐	www.marija-epita-lab.cloud-ip.cc	CNAME	ec2-13-60-230-94.eu-north-1.compute.amazonaws.com	1h	2026-02-20	edit copy delete
☐	_e8388b420b225ec6238249952766c345.marija-epita-lab.cloud-ip.cc	CNAME	9ce6735e917c811832dacaf4d0abf86b7564567	1h	2026-02-23	edit copy delete
Copy the selected records to - Select domain - execute + Add new record						

Figure 22: CName verification setting on DNS Zone (Huizhe Su)

Then we need to download and upload the certificate to our server, with following configuration. We use `scp` to upload the files to our remote instance.

```
scp ./www.marija-epita-lab.cloud-ip.cc.zip aws_epita:/home/ubuntu
```

Then we will unzip it and install them into `/etc/apache2/ssl/`

```
ip-172-31-] tmux /e/a/ssl
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example t
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf

# SSL Engine Switch:
# Enable/Disable SSL for this virtual host.
SSLEngine on

# A self-signed (snakeoil) certificate can be created by installin
# the ssl-cert package. See
# /usr/share/doc/apache2/README.Debian.gz for more info.
# If both key and certificate are stored in the same file, only th

# SSLCertificateFile directive is needed.
SSLCertificateFile /etc/apache2/ssl/certificate.crt
SSLCertificateKeyFile /etc/apache2/ssl/private.key

# Server Certificate Chain:
# Point SSLCertificateChainFile at a file containing the
# concatenation of PEM encoded CA certificates which form the
# certificate chain for the server certificate. Alternatively
# the referenced file can be the same as SSLCertificateFile
# when the CA certificates are directly appended to the server
# certificate for convinience.
SSLCertificateChainFile /etc/apache2/ssl/ca_bundle.crt

# Certificate Authority (CA):
```

Figure 23: ZeroSSL configuration (Huizhe Su)

Then, we restart apache service.

```
sudo systemctl restart apache2
```

Then we can verify the installation on ZeroSSL website.

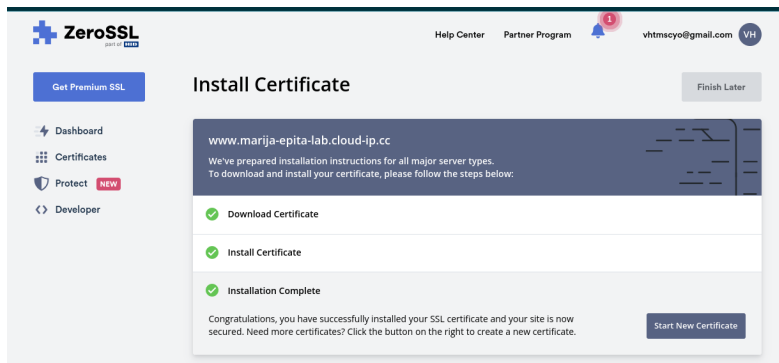


Figure 24: Verification result on ZeroSSL (Huizhe Su)

Now when we can go to our website `www.marija-epita-lab.cloud-ip.cc`, we can see the certificate is ZeroSSL.

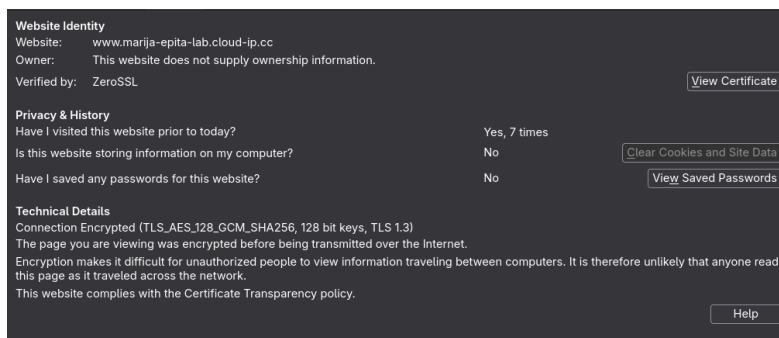


Figure 25: ZeroSSL certificate (Huizhe Su)

Question 12. Create and Install a Certified Certificate with Let's Encrypt and Certbot. Test with a Web Browser

To obtain a trusted SSL certificate with automatic renewal, we used Certbot with Let's Encrypt. First, we install Certbot and its Apache plugin:

```
sudo apt install certbot python3-certbot-apache
```

Then we request a free SSL certificate for our domain:

```
sudo certbot --apache -d arihant-dev-sail.cloud-ip.cc
```

Certbot automatically configures Apache to use the new certificate.

```

arihant@ip-172-31-37-82: /etc/apache2/ssl - ssh - /Documents/epita/masters_computer_science/big_data_infrastructure...
arihant@ip-172-31-37-82: /etc/apache2/ssl$ sudo certbot --apache -d ec2.arihant-dev-sail.cloud-ip.cc
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Enter email address (used for urgent renewal and security notices)
(Enter 'c' to cancel): arihant.jain@epita.fr

-----
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.4-August-18-2025.pdf. You must agree
in order to register with the ACME server. Do you agree?
(Y/yes/N/no): y

-----
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
(Y/yes/N/no): n

Account registered.
Requesting a certificate for ec2.arihant-dev-sail.cloud-ip.cc

Successfully resolved certificate.
Certificate is saved at: /etc/letsencrypt/live/ec2.arihant-dev-sail.cloud-ip.cc/fullchain.pem
Key is saved at: /etc/letsencrypt/live/ec2.arihant-dev-sail.cloud-ip.cc/privatekey.pem
This certificate expires on 2026-05-29.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
We were unable to find a vhost with a ServerName or Address of ec2.arihant-dev-sail.cloud-ip.cc.
Which virtual host would you like to choose?
-----
1: 000-default.conf | | Enabled
2: default-ssl.conf | | HTTPS | Enabled
-----
Select the appropriate number [1-2] then [Enter] (press 'c' to cancel): 2
Successfully deployed certificate for ec2.arihant-dev-sail.cloud-ip.cc to /etc/apache2/sites-enabled/default-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://ec2.arihant-dev-sail.cloud-ip.cc

```

Figure 26: Certbot successfully obtaining a Let's Encrypt certificate (Arihant Jain)

The site is now accessible securely at <https://arihant-dev-sail.cloud-ip.cc> with a valid SSL certificate issued by Let's Encrypt. The browser shows a padlock icon confirming the secure connection.

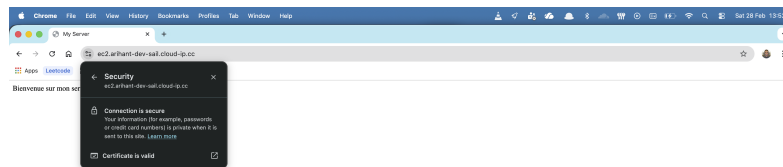


Figure 27: Valid Let's Encrypt certificate in the browser (Arihant Jain)

Question 13. Enable the Apache Rewrite Engine and Create the HTTP to HTTPS Redirect Rule. Test with a Web Browser

To ensure all traffic is served over HTTPS, we enable the Apache rewrite module and configure an automatic redirect from HTTP to HTTPS:

```

sudo a2enmod rewrite
sudo systemctl restart apache2

```

We then edit the Apache virtual host configuration for port 80:

```

sudo nano /etc/apache2/sites-available/000-default.conf

```

Adding the following rewrite rules inside the <VirtualHost *:80> block:

```

<VirtualHost *:80>
    ServerName arihant-dev-sail.cloud-ip.cc
    RewriteEngine On
    RewriteCond %{HTTPS} off

```

```
RewriteRule ^(.*)$ https://%{HTTP_HOST}%{REQUEST_URI} [R=301,L]
</VirtualHost>
```

After restarting Apache, any request to `http://arihant-dev-sail.cloud-ip.cc` is automatically redirected to `https://arihant-dev-sail.cloud-ip.cc`. Note that Certbot can also configure this redirect automatically when prompted during certificate installation.

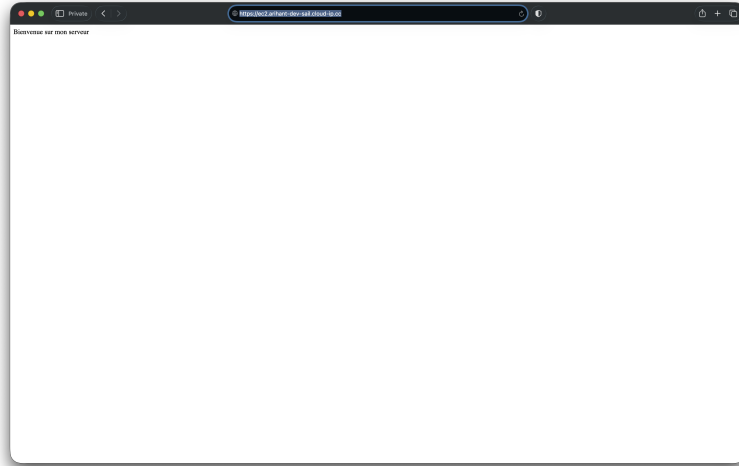


Figure 28: HTTP to HTTPS redirect working in the browser (Arihant Jain)

Exercise 3. Creating a PHP Application

Question 14. Install the PHP Module

To serve dynamic content with Apache, we install PHP and the Apache PHP module:

```
sudo apt install php libapache2-mod-php php-curl
sudo systemctl restart apache2
```

PHP is now installed and loaded as an Apache module, allowing the server to process `.php` files.

Question 15. Create a `phpinfo.php` Script and Identify the PHP Version

To verify that PHP is correctly installed and configured with Apache, we create a test file:

```
sudo nano /var/www/html/phpinfo.php
```

With the following content:

```
<?php
phpinfo();
?>
```

Navigating to `https://arihant-dev-sail.cloud-ip.cc/phpinfo.php` displays the full PHP configuration page. From this page, we can identify the PHP version currently installed on the server.



Figure 29: PHP info page showing the installed PHP version (Arihant Jain)

Question 16. Create a meteo.php Script That Displays Weather Information Using Open-Weather

To build a weather application, we first set up an OpenWeatherMap API key:

1. Create an account on <https://openweathermap.org>
2. Navigate to the API keys section and generate a new key
3. The API endpoint used is:

`http://api.openweathermap.org/data/2.5/weather?q={city}&units=metric&lang=fr&appid={apiKey}`

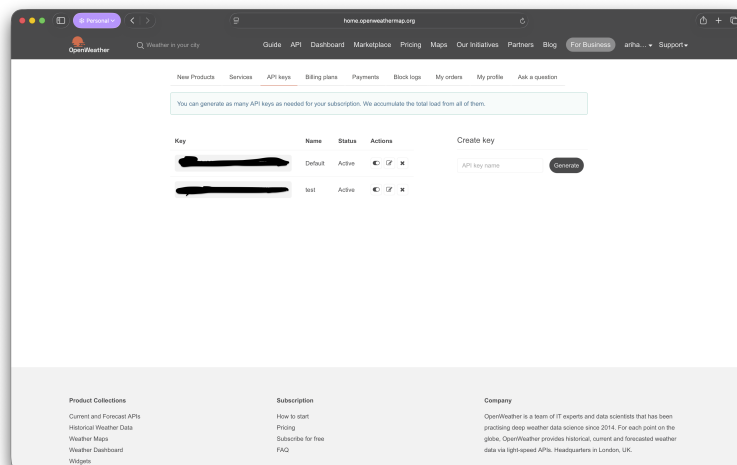


Figure 30: OpenWeatherMap API key dashboard (Arihant Jain)

We then create the weather application file:

```
sudo nano /var/www/html/meteo.php
```

The application fetches weather data for Paris using `file_get_contents()` to call the API and `json_decode()` to parse the JSON response:

```
<?php
$apiKey = "YOUR_API_KEY";
$city = "Paris";

$url = "http://api.openweathermap.org/data/2.5/weather?q="
    . $city . "&units=metric&lang=fr&appid=" . $apiKey;

$response = file_get_contents($url);
$data = json_decode($response, true);

$temperature = $data['main']['temp'];
$description = $data['weather'][0]['description'];
$humidity = $data['main']['humidity'];
$windSpeed = $data['wind']['speed'];
?>

<!DOCTYPE html>
<html>
<head>
    <title>Meteo Paris</title>
</head>
<body>
    <h1>Meteo a <?php echo $city; ?></h1>
    <p>Temperature : <?php echo $temperature; ?> C</p>
    <p>Description : <?php echo $description; ?></p>
    <p>Humidite : <?php echo $humidity; ?>%</p>
    <p>Vent : <?php echo $windSpeed; ?> m/s</p>
</body>
</html>
```

The application is accessible at <https://arihant-dev-sail.cloud-ip.cc/meteo.php>. It calls the OpenWeatherMap API with the city set to Paris, retrieves the current temperature, weather description, humidity, and wind speed, and displays them on the page. The `units=metric` parameter ensures temperatures are in Celsius and `lang=fr` returns descriptions in French.

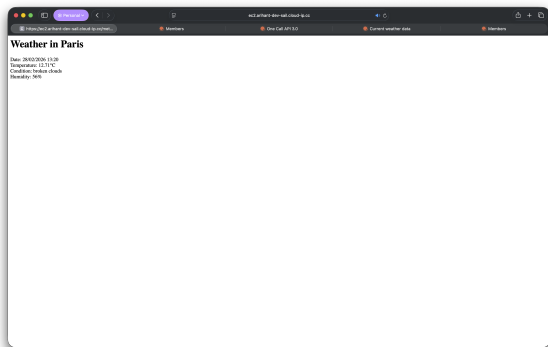


Figure 31: Initial weather page (Arihant Jain)

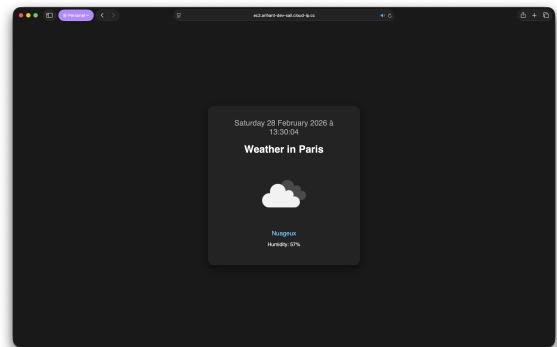


Figure 32: Updated weather page (Arihant Jain)

The left figure shows the initial version of the weather application displaying raw data from the API. The right figure shows the updated version with improved styling and layout for a better user experience.